

LOCAL EXTREMA OF CENTRAL HYPERPLANE SECTIONS OF THE FIVE-DIMENSIONAL CUBE

ZIJIAN ZENG

ABSTRACT. For a unit normal $v \in S^4$, let $\sigma(v)$ be the four-dimensional volume of the central section of $[-1/2, 1/2]^5$ perpendicular to v . We classify all local extrema of σ . Up to signed coordinate permutations, they are precisely the one-, two-, and five-diagonal directions. The first is a local minimum, and the latter two are local maxima. We also classify the full-support critical set: besides the five-diagonal, it contains exactly one orbit, $(\alpha, \alpha, 1, 1, 1)$ with $\alpha = (24 + \sqrt{69})/13$, and this orbit is a saddle. The proof reduces the positive ordered cone to six pair-sum chambers. Each chamber and all incident walls are closed by exact rational certificates: Bernstein positivity, saturated ideal decompositions, and Sturm isolation. Independent verifiers reconstruct the polynomial systems; their canonical JSON loaders reject duplicate object keys, booleans in integer slots, and targeted semantic corruptions.

1. INTRODUCTION

Let

$$Q_n = [-1/2, 1/2]^n, \quad \sigma_n(v) = \text{Vol}_{n-1}(Q_n \cap v^\perp), \quad v \in S^{n-1}.$$

A direction is *diagonal* if its nonzero coordinate absolute values are equal. Thus its orbit under signed coordinate permutations is represented by

$$d_k = (\underbrace{1, \dots, 1}_k, 0, \dots, 0)/\sqrt{k}.$$

Ambrus and Gárgyán conjectured that, for every $n \geq 2$, every locally extremal central section of Q_n is diagonal [2, Conjecture 1.3]. They also constructed non-diagonal critical sections in every dimension at least four and proved that their selected family consists of saddles. The four-cube critical directions had previously been classified by Ambrus [1]; the sole non-diagonal orbit is represented by $(1, 1, 2, 2)/\sqrt{10}$.

The status of diagonal sections itself requires care. Ambrus and Gárgyán proved that, for $n \geq 4$, d_k is not locally extremal when $3 \leq k \leq n-1$ [3].

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Pournin’s erratum corrects an omitted transverse term in the constrained-Hessian criterion for proper subdiagonals and explicitly leaves the main-diagonal conclusions unchanged [5]. The sharp global upper bound is Ball’s cube slicing theorem [4].

The database-bounded literature search recorded with this manuscript found no prior complete classification in dimension five. In that precise sense, our main result addresses the first dimension not covered by the preceding classifications.

Theorem 1.1. *Up to signed coordinate permutations, the locally extremal directions of Q_5 are exactly*

$$d_1 = (1, 0, 0, 0, 0), \quad d_2 = (1, 1, 0, 0, 0)/\sqrt{2}, \quad d_5 = (1, 1, 1, 1, 1)/\sqrt{5}.$$

The direction d_1 is a strict local minimum, d_2 is a global maximum, and d_5 is a strict local maximum. Consequently every locally extremal central hyperplane section of Q_5 is diagonal.

The computational part of the proof is exact rather than numerical. All coefficients lie in $\mathbb{Z}$ or $\mathbb{Q}$; the only algebraic number in the final classification is isolated by a quadratic polynomial and rational endpoints. The release archive contains serialized certificates, independent verifiers that reconstruct their source equations, and mutation tests. No floating-point root is used to establish existence, completeness, or a Hessian sign.

2. THE SECTION FORMULA AND CRITICAL EQUATIONS

We begin with a normalization valid on every support stratum.

Lemma 2.1 (truncated-power formula). *Let $a = (a_1, \dots, a_s) \in \mathbb{R}_{\geq 0}^s$, put $R = \|a\|_2$, $A = \sum_i a_i$, and $a_E = \sum_{i \in E} a_i$. The central section volume with normal a/R is*

$$(1) \quad F_s(a) = \frac{R}{(s-1)! \prod_i a_i} \sum_{E \subseteq [s]} (-1)^{|E|} \left(\frac{A}{2} - a_E \right)_+^{s-1}.$$

If the normal is embedded in a higher-dimensional cube by adjoining zero coordinates, its section volume remains $F_s(a)$.

Proof. Let U_i be independent uniform random variables on $[-1/2, 1/2]$ and $Z = \sum_i a_i U_i$. Coarea gives

$$f_Z(t) = R^{-1} \text{Vol}_{s-1} \{x \in Q_s : a \cdot x = t\}.$$

After translating, $Y_i = a_i(U_i + 1/2)$ is uniform on $[0, a_i]$. Inclusion–exclusion on the simplex yields

$$\Pr\left(\sum_i Y_i \leq x\right) = \frac{1}{s! \prod_i a_i} \sum_{E \subseteq [s]} (-1)^{|E|} (x - a_E)_+^s.$$

Differentiate and evaluate at $x = A/2$. Zero coordinates factor off a unit cube, proving the last statement. $\square$

For $s = 5$, pair a subset with its complement. Write

$$T = \frac{1}{2} \sum_{i=0}^4 a_i, \quad S = \sum_{i=0}^4 a_i^2, \quad L_{ij} = T - a_i - a_j,$$

and $\psi(x) = x|x|^3$. Then

$$(2) \quad P = T^4 - \sum_i (T - a_i)^4 + \sum_{i < j} \psi(L_{ij}), \quad F_5(a) = \frac{\sqrt{S} P}{24 \prod_i a_i}.$$

The function ψ is C^3 , so (2) and its first three derivatives agree across every pair-sum wall.

In a strict chamber define $N = \{ij : a_i + a_j > T\}$. There the polynomial numerator is

$$(3) \quad P_N = T^4 - \sum_i (T - a_i)^4 + \sum_{i < j} L_{ij}^4 - 2 \sum_{ij \in N} L_{ij}^4.$$

If $P_i = \partial_i P$, homogeneity shows that a positive point is critical exactly when

$$(4) \quad G_i = a_i S P_i + (a_i^2 - S) P = 0, \quad 0 \leq i \leq 4.$$

One equation is dependent. No wall factor has been divided from (4).

If $a_0 \geq \sum_{i>0} a_i$, direct parametrization gives $F_5(a) = \sqrt{S}/a_0$. Its logarithmic derivative with respect to any positive a_j , $j > 0$, is $a_j/S > 0$. Hence the dominant region and its equality wall contain no full-support critical point.

3. SIX ORDERED CHAMBERS

Signed coordinate permutations allow us to assume $a_0 \geq a_1 \geq \dots \geq a_4 > 0$. In the strictly balanced cone, the graph N is shifted: replacing an endpoint by a lower index cannot decrease its pair sum. It is also intersecting, since two disjoint high pairs would sum to more than $2T = \sum_i a_i$, leaving a positive fifth coordinate.

Lemma 3.1. *The strict high-pair graph is one of*

$$(5) \quad \emptyset, \quad \{01\}, \quad \{01, 02\}, \quad \{01, 02, 03\}, \quad \{01, 02, 03, 04\}, \quad \{01, 02, 12\}.$$

Every full-support pair-wall point belongs to the closure of at least one of these chambers.

Proof. A shifted intersecting family of two-subsets is either an initial star or the triangle on $\{0, 1, 2\}$, giving (5). For the wall statement, take a generic sequence in the positive balanced cone converging to the point. After passing to a subsequence and resolving equal-coordinate order by a permutation, its strict sign graph is constant and belongs to (5). $\square$

We call the six cases empty, star1, star2, star3, star4, and triangle.

Theorem 3.2 (full-support critical set). *Up to signed coordinate permutations, the full-support critical directions of Q_5 are exactly d_5 and*

$$(6) \quad v_\alpha = \frac{(\alpha, \alpha, 1, 1, 1)}{\sqrt{2\alpha^2 + 3}}, \quad \alpha = \frac{24 + \sqrt{69}}{13}.$$

The tangent Hessian at v_α has signature $(1, 3)$ and is indefinite.

We prove Theorem 3.2 through the exact chamber propositions below. Long but purely polynomial identities are bound to the certificates described in Section 5.

3.1. The empty chamber. In the empty chamber the symmetric polynomial P can be written, with $p_j = \sum_i a_i^j$, as

$$P = \frac{3}{8}p_1^4 - 3p_1^2p_2 + 3p_2^2 + 4p_1p_3 - 4p_4.$$

At a critical point every coordinate is a root of one common nonzero quartic

$$q(X) = -16p_2X^4 + 12p_1p_2X^3 + (C_1p_2 + P)X^2 + C_0p_2X - p_2P,$$

where $C_0 = \frac{3}{2}p_1^3 - 6p_1p_2 + 4p_3$ and $C_1 = -6p_1^2 + 12p_2$. Thus there are at most four distinct coordinate values. The exact certificate exhausts the one-, two-, three-, and four-value multiplicity patterns. Divided differences and resultants reduce each branch to univariate factors; rational Sturm chains exclude every factor on its order-and-chamber interval. The four-value case is closed by Vieta relations. The sole admissible point is $a_0 = \dots = a_4$.

3.2. The star1 chamber. Normalize $a_4 = 1$ and introduce ordered gaps

$$(a_0, \dots, a_4) = (1 + w + z + y + x, 1 + w + z + y, 1 + w + z, 1 + w, 1).$$

Putting $y = (1 - x - z)/2 + u$ identifies the entire ordered closure with

$$(7) \quad x, z, u, w \geq 0, \quad x + z \leq 1.$$

For a gap variable $r \in \{x, y, z, w\}$ let

$$H_r = S_rPD + 2SP_rD - 2SPD_r, \quad D = \prod_i a_i.$$

The degree-nine simplex-Bernstein expansion of

$$(8) \quad K = -6H_x + 7H_y - 8H_z + 4H_w$$

has zero coefficient only at the $(0, 0)$ vertex. Every monomial coefficient in each of the other 54 parameter polynomials is strictly positive, including its constant coefficient. Hence $K = 0$ on (7) forces $x = z = 0$.

On that face two necessary equations reduce, after positive factors are removed, to polynomials $A(u, w), B(u, w)$. Their exact radical decomposition has five graph components, whose univariate degrees are 2, 2, 2, 3, 6. Four components contradict $u, w \geq 0$. The surviving component is

$$w = 0, \quad 52u^2 - 36u - 15 = 0.$$

Its unique nonnegative root is $u = (9 + 2\sqrt{69})/26$ and gives precisely (6). Direct reduction verifies all four original equations, not merely the two necessary face equations.

3.3. The star2 chamber. The ordered closure is parametrized by

$$x = u + v, \quad z = u + 1 - v, \quad u, y, w \geq 0, \quad 0 \leq v \leq 1.$$

For $H = -2H_x + 2H_y - 2H_z + H_w$, exact conversion to the degree-eight Bernstein basis in v gives

$$H = \sum_{k=0}^8 B_k(u, y, w) \binom{8}{k} v^k (1-v)^{8-k}.$$

Each B_k has 125 monomials and all 1125 rational coefficients are strictly positive; the least coefficient is 32. Thus $H > 0$ on the entire closure, contradicting simultaneous criticality. A separate degree-four Bernstein table has 70 positive coefficients and certifies $P > 0$.

3.4. The star3, star4, and triangle chambers. For star3, saturation of the four equations by $a_0 a_1 a_2 a_3 PS$ yields a radical degree-14 ideal. It decomposes into four components of degrees 2, 4, 4, 4. With $q(s) = 2s^4 + 5s^2 - 1$, the quadratic component is parametrized by $(2t, t, t, t)$, $3t^2 - 1 = 0$, while the three quartic components are coordinate permutations of

$$(s + \ell, \ell, s, s), \quad \ell = \frac{s^2 + 1}{2s}, \quad q(s) = 0.$$

Exact real-root isolators show $0 < t < 2/3$ and $0 < s < 1/2$ on the positive branches. Both violate the normalization order $a_3 \geq a_4 = 1$.

For star4, six certified low-degree ideal consequences force equality patterns among the four leaves. Every unequal-leaf branch contradicts the smallest high-edge margin. The all-equal branch reduces to

$$R(X) = X^6 - 12X^5 + 51X^4 - 96X^3 + 96X^2 - 64.$$

A primitive rational Sturm chain has two variations both at $X = 2$ and at $+\infty$, so there is no root in $(2, \infty)$. The endpoint is not a root. Every pair-wall boundary of star4 belongs to a shorter-star closure.

For the triangle, the saturated ideal is one-dimensional. An exact Groebner basis contains

$$(a_2^2 - a_3^2 - 1)(a_2^2 - 4a_3^2 - 4).$$

The remaining basis equations first force $a_0 = a_1$, and then $a_0 = a_2$ or $a_0 = 2a_2$; either branch forces $a_2^2 - a_3^2 - 1 = 0$. The other factor cannot vanish simultaneously over $\mathbb{R}$. But the weak high-edge inequality gives $a_2 \geq a_3 + 1$, whereas

$$(a_3 + 1)^2 - a_2^2 = 2a_3 > 0,$$

a contradiction. Since only weak inequalities were used, every incident wall is included.

3.5. Hessian at the surviving non-diagonal point. At (6), the tangent space splits under $S_2 \times S_3$ into a large-block difference, two small-block differences, and the block-contrast line. Reduction modulo

$$(9) \quad 13\alpha^2 - 48\alpha + 39 = 0, \quad \frac{12}{5} < \alpha < \frac{5}{2},$$

gives, up to positive denominators, the following quadratic forms:

$$-4(17784\alpha + 13597), \quad 4(416\alpha - 1367), \quad 12(104\alpha - 215).$$

The first two are negative and the last is positive on the isolating interval. Their multiplicities are 1, 2, 1, respectively. At d_5 , direct differentiation of (2) gives four tangent eigenvalues

$$(10) \quad -\frac{5\sqrt{5}}{32} < 0.$$

This completes the proof of Theorem 3.2.

4. SUPPORT STRATA AND PROOF OF THE MAIN THEOREM

If a point of S^4 is a local extremum, its restriction to the coordinate subsphere determined by its support is a local extremum of the corresponding F_s . It therefore suffices to classify the lower supports and test the remaining diagonal points transversely.

For $s = 3$, the dominant region has no positive critical point. In the balanced region normalize the ordered coordinates to $(a, b, 1)$. Let $P = T^2 - \sum_{q \in \{a, b, 1\}} (T - q)^2$, $D = ab$, and form the two cleared derivatives H_a, H_b . After the positive saturation $H_a/b, H_b/a$, exact elimination gives

$$(11) \quad \text{Res}_a(H_a/b, H_b/a) = -64b^2(b-1)^6(b^2+b+1)^3.$$

Positivity forces $b = 1$; then the two equations are $-2a^2(a-1)^2$ and $a^3(a-1)^2$, so $a = 1$. The two-dimensional case is immediate from $F_2 = \sqrt{S}/a_0$ off the diagonal.

For $s = 4$, exact reconstruction of the classification in [1] gives only the diagonals and

$$(12) \quad p = (1, 1, 2, 2)/\sqrt{10}.$$

The tangent eigenvalues at p are

$$(13) \quad -\frac{5\sqrt{10}}{12}, \quad -\frac{25\sqrt{10}}{24}, \quad \frac{5\sqrt{10}}{24},$$

so its embedding in S^4 is a saddle.

Near d_1 , the largest coordinate dominates and, on the unit sphere, $F_5 = 1/a_0 \geq 1$, with equality only at d_1 in its orbit. Thus d_1 is a strict local minimum. Ball's theorem makes d_2 a global maximum [4]. At d_3 the support-three path $(1+u, 1+u, 1-2u)$ has

$$(14) \quad F_3 = \frac{3\sqrt{3}(2u+1)\sqrt{2u^2+1}}{4(1+u)^2} = \frac{3\sqrt{3}}{4} + \frac{3\sqrt{3}}{2}u^3 + O(u^4),$$

so d_3 is a saddle. At d_4 , the support tangent Hessian is negative definite, whereas the transverse path $(1, 1, 1, 1, t)$, $t > 0$, satisfies

$$(15) \quad F_5 = \frac{\sqrt{t^2 + 4}(3t^3 - 16t^2 + 128)}{192} = \frac{4}{3} + \frac{t^3}{32} + O(t^4).$$

Hence d_4 is a saddle. Finally (10) makes d_5 a strict local maximum. Together with Theorem 3.2, this proves Theorem 1.1.

5. EXACT CERTIFICATES AND INDEPENDENT VERIFICATION

The proof archive separates discovery programs from release verifiers. Every decisive verifier reconstructs the relevant instance of (3)–(4) from serialized chamber data. Unknown or missing fields, unavailable exact backends, timeouts, nonzero normal forms, malformed success sentinels, duplicate JSON keys, and noncanonical booleans are rejected. No verifier imports a discovery module or a cached numerical root list.

Endpoint	SHA-256 prefix	Exact method
Q_3 support	6f704264	resultant and positivity
Q_4 non-diagonal	1c732a3b	rational tangent form
empty	3d5c97a2	resultants, Vieta, Sturm
star1	70cf7f98	Bernstein and radical decomposition
star2	fd591e36	strict Bernstein positivity
star3	f966e220	radical decomposition and isolation
star4	b57cad6a	ideal consequences and Sturm
triangle	5e0bab27	saturated basis and inequalities

TABLE 1. Decisive exact certificates. The manifest records complete hashes.

From the project root, the complete audit is reproduced by
`/opt/anaconda3/bin/python3 src/verify_all.py`

This runs formula reconstruction, all support and chamber verifiers, independent checks, and targeted corrupted-input tests. The master record is written to `audit/verification_log.json`. Several verifiers use only the Python standard library; ideal computations invoke Singular over $\mathbb{Q}$, and the remaining exact reconstructions use SymPy.

COMPUTATIONAL AND AI-ASSISTANCE DISCLOSURE

Symbolic algebra, exact polynomial scripts, automated searches, and language models were used during discovery, proof organization, code drafting, and adversarial test generation. Every theorem endpoint is instead bound to exact serialized input and independently reconstructing verifiers as described above. Numerical searches are not proof inputs. No Lean, Coq, Isabelle, or other interactive proof assistant was used.

DATA AND CODE AVAILABILITY

The source archive accompanying this manuscript contains all certificates, verifiers, mutation tests, the master verification log, the literature audit, and the complete $\text{\LaTeX}$ source. The certificate manifest fixes every released file by SHA-256.

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INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR, 56000, MALAYSIA

Email address: `zijianzeng@foxmail.com`

Email address: `1002266693@ucsiuniversity.edu.my`